Beyond Magnitude: Attribution-Guided Structural Pruning of Mamba State Space Models

Chirag (DA25M008)

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Code: github.com/chirag3211/mamba-attribution-pruning

1 Introduction

Selective state space models such as Mamba [1] are a linear-time alternative to Transformers for language modelling. To make a pretrained Mamba cheaper to run, I study **structured pruning**: removing whole inner channels of every Mamba block, which shrinks the input projection, the depthwise convolution, the SSM parameters and the output projection together and gives a smaller dense model on ordinary hardware. The open question is the **importance score** that decides which channels to remove. The usual choice, weight magnitude [5, 6], ignores that every Mamba channel passes through a multiplicative gate before it reaches the output, so its influence depends on activations and not only on weights.

I ask whether attribution scores from MambaLRP [2], a layer-wise relevance propagation (LRP [3]) method adapted to Mamba, choose better channels than magnitude, gradient-based and random scores, and **which ingredient of an attribution score does the work**. My contributions are:

- a tested multi-seed grid for channel pruning of Mamba with 16 selection methods, 147 runs on mamba-130m and 75 on mamba-370m;
- paired statistics with confidence intervals instead of single runs;
- the findings that seeing the gate half of the input projection is the decisive ingredient, that weight magnitude is worse than random, and that LRP's special propagation rules add little beyond simpler gate-aware scores.

2 Prior Work

Mamba [1] computes, in each block, an input projection into a content half and a gate half; the content passes through a causal convolution and a selective SSM and is multiplied by SiLU(gate). **MambaLRP** [2] makes LRP faithful for this architecture by adding rules for the SiLU gate, the SSM recurrence and normalisation, and uses it to explain predictions; it has not been used to prune. **Pruning by explaining** [4] showed for CNNs that LRP relevance is a strong pruning criterion, while magnitude pruning [5, 6], first-order Taylor/gradient criteria [7] and mean-activation criteria [8] are the standard baselines. I build on [2] and [4] and compare against these baselines on Mamba, using WikiText-103 [9] for calibration, recovery and evaluation.

3 Idea and Implementation

Pruning operation. Removing channel c of a block deletes rows c and $c{+}D$ of the input projection, the convolution channel, the matching input column of `x_proj`, row c of `dt_proj`, the input column of the output projection, and entry c of `A_log` and `D`. Every layer keeps the same number of channels (per-layer top-k, widths rounded to multiples of 8 for the fused kernels), so all methods give models of identical size: mamba-130m (24 layers, 1,536 channels) keeps 69.8/50.0/30.2% of its channels at 30/50/70% removal (1,072/768/464 of 1,536), and mamba-370m (48 layers, 2,048 channels) is pruned to the same fractions.

Scores. My idea is to score each channel by the relevance that reaches it through **both** halves of the input projection, the content path and the gate path. I compare 16 selections (Table 1): LRP at three sites (LRP-CO content half only; LRP-GI content plus gate half, i.e. the proposed score; LRP-Out at

the post-gate site); plain gradient×input and Taylor $|w\,\partial L/\partial w|$, which use the same sites without LRP's rules; mean absolute activation, which needs no backward pass; weight magnitude; Random; and two hybrids that use the gate-aware score only in layer 0 or everywhere except layer 0. Scores are computed on 256 calibration sequences of 128 tokens (MambaLRP on its slow PyTorch path; fused kernels afterwards).

Table 1: The channel-importance scores compared. "-CO" = content half only, "-GI" = gate included, "-Out" = post-gate site.

Family	Variants	What is measured	Cost
LRP [2]	LRP-CO, LRP-GI, LRP-Out	relevance reaching the channel through the content half, content+gate halves, or the post-gate site	forward + relevance backward
Grad×input	GxI-CO, GxI-GI, GxA-Out	plain gradient times input at the same sites (no LRP rules)	forward + backward
Taylor [7]	GradMag-CO, GradMag-GI	$\lvert w\,\partial L/\partial w\rvert$ on the input-projection rows	forward + backward
Activation [8]	Act-CO, Act-GI, Act-Out	mean |activation| at the same sites	forward only
Magnitude [5, 6]	Mag-CO, Mag-GI	mean $\lvert w\rvert$ of the content rows, or content+gate rows	none (weights only)
Random; hybrids	Random, Hyb-L0GI, Hyb-L0CO	random ranking; LRP-CO with the LRP-GI ranking in layer 0 (Hyb-L0GI) or the reverse (Hyb-L0CO)	as LRP

Protocol. After pruning, all weights are fine-tuned for 500 steps (AdamW, lr $2 \cdot 10^{-4}$, 20 warm-up steps, cosine decay, batch 8×128 tokens) and evaluated on the full WikiText-103 test set (286,208 tokens in 256-token chunks). Three seeds change calibration data, fine-tuning order and the random score. Differences are **paired**: per seed I form Δ log-perplexity between two methods, report the percentage change with a 95% t-interval over seeds ($n = 3$) and the number of seeds in which A wins; "equivalent" means the interval lies within ±2%. Contrasts C1–C8 were fixed before the grid ran; C9–C18 were added after seeing seed 0 or the overlap analysis and are exploratory. The unit tests check that my scores equal the original formulas (relative error 0) and that a pruned model equals the zeroed model.

Implementation and reproducibility. `mamba_grid.py` runs the whole grid (score → prune → fine-tune → evaluate) with fused Mamba kernels on a single T4 GPU, caches scores per seed, skips finished runs and writes one JSON per run; `aggregate.py` produces every table, contrast and figure here; `compare_pruned_models.py` rebuilds the kept-channel sets from the cached scores without a GPU. A single Colab notebook (`grid_all.ipynb`) drives it end to end, one account per seed; a run takes 5–14 minutes. The README of the repository gives the exact commands.

4 Evaluation

Unpruned perplexity is 32.44 (130M) and 22.42 (370M). Fine-tuning alone improves the dense model to 22.17 and 17.71, which is the right reference for pruned models. Table 2 gives post-fine-tuning perplexity; Table 3 gives paired contrasts (negative = first method better).

Table 2: Post-fine-tuning WikiText-103 perplexity, geometric mean over 3 seeds (bold = best). "–" = not run at that size. Seed spread (sd of log) ≤ 0.02 except Random and the magnitude baselines.

	mamba-130m, % removed			mamba-370m, % removed		
Method	30%	50%	70%	30%	50%	70%
Dense, same fine-tuning	22.2			17.7		
Random	41.3	76.0	216.5	30.0	50.6	117.2
Mag-CO	57.3	117.7	302.0	–	–	–
Mag-GI	62.3	143.4	469.8	–	–	–
GradMag-CO	35.2	59.2	124.8	–	–	–
GradMag-GI	31.6	51.7	98.5	23.6	36.7	74.1
Act-CO	40.9	72.2	136.7	–	–	–
Act-GI	37.4	59.7	114.9	–	–	–
Act-Out	31.0	45.6	83.7	23.5	34.1	62.8
GxI-CO	34.2	55.6	111.2	–	–	–
GxI-GI	**29.9**	46.7	**82.7**	**22.7**	**32.7**	**61.6**
GxA-Out	30.6	47.0	86.1	–	–	–
LRP-CO	37.5	60.5	107.7	25.0	37.6	73.4
LRP-GI	30.4	**45.5**	86.0	23.3	33.7	64.4
LRP-Out	30.4	45.9	85.6	–	–	–
Hyb-L0GI	35.1	54.4	94.4	24.7	36.4	70.4
Hyb-L0CO	32.1	50.3	97.3	23.6	34.7	67.1

Table 3: Paired contrasts: change in post-fine-tuning perplexity (%) of A relative to B. All differences are in the same direction in 3 of 3 seeds and have 95% intervals excluding 0, except those marked *. Full intervals, one-shot numbers and all 18 contrasts are in the repository.

	mamba-130m			mamba-370m		
Contrast (A vs B)	30%	50%	70%	30%	50%	70%
C1 Gate path: LRP-GI vs LRP-CO	−19.1	−24.8	−20.2	−7.1	−10.3	−12.3
C8 LRP-GI vs Random	−26.4	−40.0	−60.3	−22.4	−33.4	−45.0
C16 LRP-GI vs Mag-GI (magnitude)	−51.3	−68.2	−81.7	–	–	–
C17 LRP-GI vs Mag-CO (magnitude)	−47.0	−61.3	−71.5	–	–	–
C4 LRP-GI vs GradMag-GI (Taylor)	−3.9	−11.8	−12.7	−1.4	−8.3	−13.1
C2 LRP-GI vs GxI-GI (plain grad×input)	+1.4	−2.4	+3.9	+2.3	+3.0	+4.6
C5 LRP-GI vs Act-Out (activation)	−2.0	−0.1*	+2.7	−0.9	−1.3	+2.5
C7 Act-Out vs Act-GI (post- vs pre-gate)	−17.2	−23.6	−27.2	–	–	–
C13 Hyb-L0GI vs LRP-CO (gate in layer 0 only)	−6.4	−10.1	−12.3	−1.5	−3.2	−4.1
C15 Hyb-L0CO vs LRP-GI (no gate in layer 0 only)	+5.7	+10.5	+13.1	+1.3	+2.9	+4.3

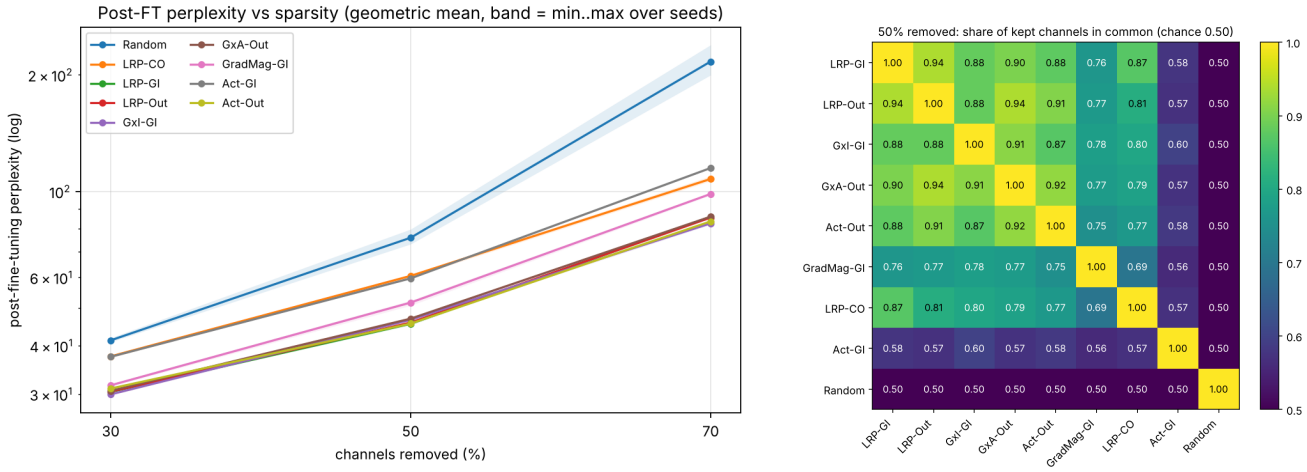


Figure 1: Left: post-fine-tuning perplexity vs. sparsity on mamba-130m (log scale; the five gate-aware scores overlap at the bottom). Right: share of kept channels two methods have in common at 50% removed (chance = 0.50).

4.1 Findings

Attribution beats random, gradient and magnitude baselines. LRP-GI is 26%, 40% and 60% better than random on 130M (22–45% on 370M) and 3.9–12.7% better than the Taylor score (1.4–13.1% on 370M), in every seed. Weight magnitude, the classic baseline, is worse than random: after fine-tuning, content-only magnitude (Mag-CO) is 39%, 55% and 40% worse than random removal, and gate-inclusive magnitude (Mag-GI) 51%, 89% and 117% worse (30/50/70% removed; 3 of 3 seeds); one-shot perplexity is 10^{10}–10^{17}. LRP-GI beats Mag-GI by 51%, 68% and 82% (C16). This matches the earlier single-seed finding that magnitude is uncorrelated with relevance (Spearman $\rho = 0.043$), and means that for Mamba the standard baseline is not merely weaker but actively misleading.

The gate path is the decisive ingredient. Adding the gate half cuts post-fine-tuning perplexity by 19–25% for LRP on 130M (7–12% on 370M) and by 8–26% for gradient×input, activation and Taylor scores (C9–C11, 130M); one-shot, before fine-tuning, the gain is 91–99% for these three families. Measuring activation after the gate beats measuring it before by 17–27% (C7).

LRP's special rules add little once the gate is included. Plain gradient×input is within 4% of LRP-GI (sign changes with sparsity on 130M; 2–5% better in every seed on 370M) and mean post-gate activation, which needs only a forward pass, is within 3% (C2, C5). The practical recommendation is therefore to score channels at the post-gate site and prefer the cheap activation score.

Why: the selections differ in few channels, mostly in layer 0. The five gate-aware scores keep 87–96% of the same channels (30–50% removed), and content-only LRP shares 91/87/83% of its kept channels with LRP-GI yet is 19–25% worse. In layers 1–23 the two agree on 74–95%, but in layer 0 only on 72/48/25%, which is chance level (70/50/30%); the same holds on 370M (69/42/31% in layer 0 vs. 88–93% elsewhere). Hybrid selections test this causally: the gate in layer 0 alone recovers 31%, 37% and 58% of the gate effect on 130M (C13; 21–32% on 370M) and removing it only in layer 0 costs 26–55% of it (C15), so layer 0 is a large but incomplete part of the effect.

Costs are real. The best 30%-pruned model is still 35% above the fine-tuned dense model on 130M (28% on 370M) and 3.7× at 70% removed, so these are accuracy–size trade-offs, not free compression.

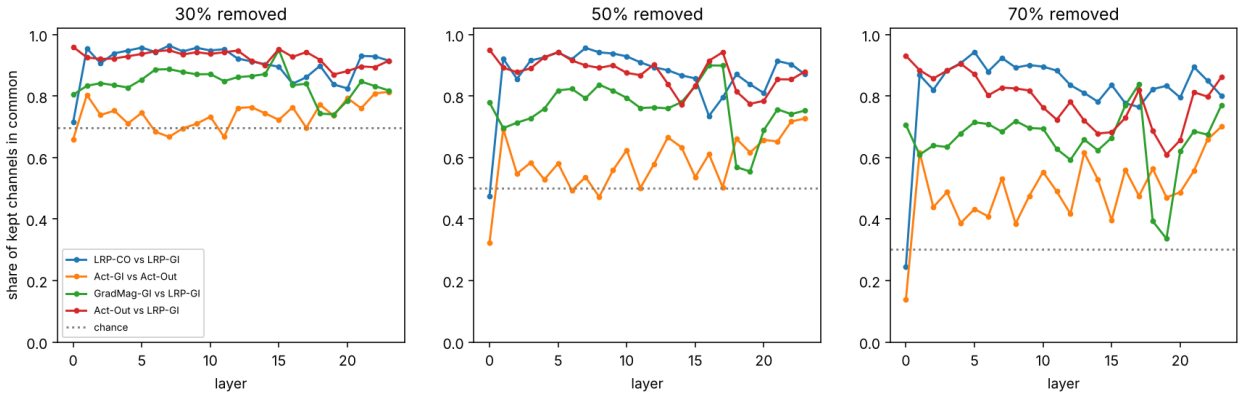


Figure 2: Per-layer share of kept channels in common (mamba-130m, mean over seeds). Content-only vs. gate-inclusive LRP (blue) and pre- vs. post-gate activation (orange) disagree at chance level in layer 0 and are otherwise far above chance for LRP.

One-shot vs. recovered quality. Before fine-tuning the scores differ by orders of magnitude (random removal at 70% reaches perplexity $3 \cdot 10^{16}$ on 130M, content-only LRP $3 \cdot 10^{4}$, LRP-GI about 10^{3}), but 500 fine-tuning steps remove most of the gap within 100 steps (Figure 3), and all methods except the magnitude baselines lie within a factor of 2.6 afterwards. One-shot perplexity is therefore a more sensitive but less relevant measure; post-fine-tuning perplexity is the primary endpoint throughout.

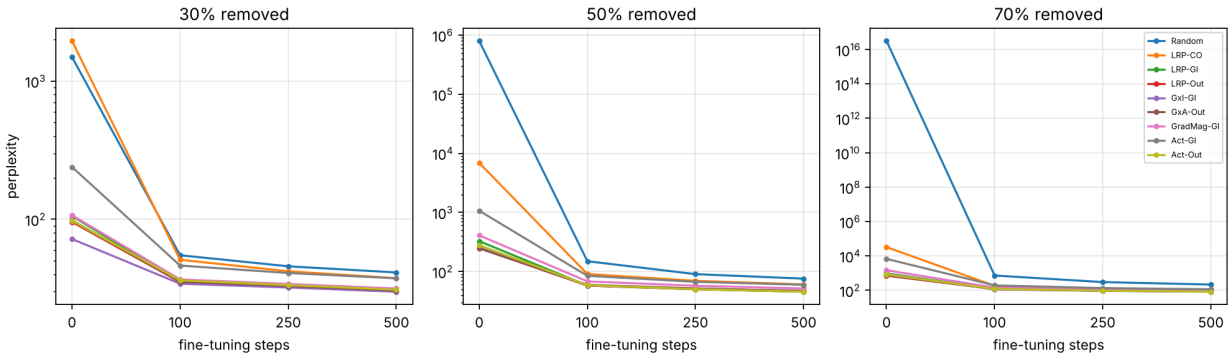


Figure 3: Perplexity after 0, 100, 250 and 500 fine-tuning steps (mamba-130m, log scale).

4.2 Limitations

Two checkpoints of one model family, three seeds (intervals rest on $n = 3$), in-domain fine-tuning with perplexity as the only metric, uniform per-layer budgets and a fixed 500-step recovery. My magnitude baselines use the mean absolute value of the input-projection rows only (content half, or content plus gate half); other definitions (e.g. norms over all weights of a channel) were not tested. Weight magnitude and the other 130M-only selections were not run on mamba-370m (8 of the 16 selections were), and C9–C18 and the hybrid test were designed after seeing earlier results, so they are hypothesis-generating. Earlier single-seed exploratory work (hypotheses H1–H15) is in the repository.

5 Conclusion

Channel importance in Mamba should be measured where the gate has already acted. Weight magnitude, the standard baseline, is worse than random removal here, while including the gate path of the input projection improves every score family and removes the apparent advantage of content-only scores. LRP's specially corrected rules add at most a few percent over plain gradient×input or activation, which costs a single forward pass. Layer 0 accounts for a large part of the gate effect. Next steps are larger Mambas, non-uniform per-layer budgets and downstream accuracy.

References

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[10] This work: code, raw per-run results and reproduction instructions: https://github.com/chirag3211/mamba-attribution-pruning